# Tutorial 6: FINA2205 Portfolio Theory and Asset Pricing 2

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# Group assignment

## Key Information

**Page limit:** 13 pages (or max 14 pages if references require 2 pages) in total, including 1-page title page, 10-page main text, 1-page Appendix (if any) and references of 1 to 2 pages.

**Weighting:** 30%

**Individual/team:** Team

**Due date:** 11:59pm AWST Monday 4th May 2026

After you have read this information, go to the **Assessment 2 Q&A** discussion board to ask any questions and see what your peers are saying about this assignment.

## Submission Requirements

Two files: one report in PDF format (not MS Word, as the pagination may change), and the Excel completed file with the template file name renamed.

Your assessment submission should:

- properly cite any sources or references used in your report using the APA Style.

Hi Yihui,

I have a question ab

In Part 1.3, you require us to obtain the daily risk-free rat
the annual rate to the daily rate. We are only seeming to
would like some clarification if possible.

Reply

Yihui Lan UNIT COORDINATOR
16 Apr 2026 2:26 pm

Hi Grace,

The 4-week Treasury Bill rate (DTB4WK) is referred t
Economic Data); the values provided are actually an

For example, on 28/2/2026, the TB rate is 4% per an
why we need to convert this annualised rate to a da
a stock. We need to convert the annual rate to matc

Please let me know if you have additional queries.

Best,
Yihui

Hi Yihui, Excel won't let me delete the rows for the dates 13/10/25 and 11/11/25 (Empty value from FRED data) as I'm using the STOCKHISTORY formula for it, so I was wondering if it's okay to hardcode the values from STOCKHISTORY then deleting the rows for the

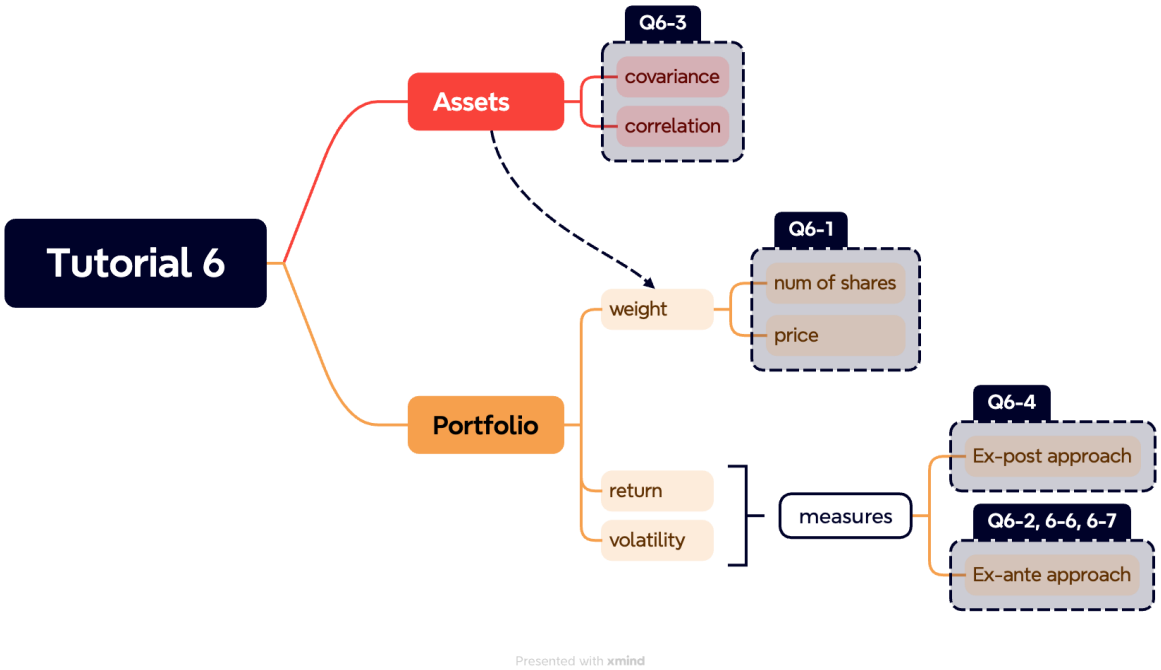
Hi, just wondering whether there is a mistake in the Excel file for the assignment. On sheet "8_Testing_dummy_coeff, there are columns for calculating the T-stat and P-value for testing: H0: y=0, H1: y!=0, but for the CAPM model WITHOUT the dummy variable included. Is this possible? If so, how can this be done?

Thanks!

| | $H_1: \gamma \neq 0$ | CAPM | | | CAPM with Dummy | | |
|---|---|---|---|---|---|---|---|
| | Asset | Stock 1 | Stock 2 | Portfolio | Stock 1 | Stock 2 | Portfolio |
| | **Beta** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **(SE)** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| **Rejection Region (RR) rule** | | | | | | | |
| | **t-stat** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **Critival velues** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **T-stat in the RR Y/N?** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **Conclusion (reject/do not reject)** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| **p-value rule** | | | | | | | |
| | **p-value (based on t-stat)** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **p<=0.05? (Y/N)** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |
| | **Conclusion (reject/do not reject)** | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD | TO ADD |

# Agenda

- 7 problems



# Recap: Covariance and correlation

- Covariance: Tells us how two stocks move relative to each other.
  - Positive (+): They tend to move together.
  - Negative (-): They tend to move in opposite directions.
- The problem: The final number is messy and hard to interpret on its own.
- Correlation ($\rho$): It is a <u>standardized version</u> of covariance.
- The Golden rule: Correlation will **ALWAYS** be between -1 and +1.
- It gives us a clean, universally understood percentage of how strongly they are linked.

# Recap: Covariance and correlation

Formula:

$$Cov(R_A, R_B) = \frac{1}{n-1}\sum_t (R_{A,t} - \overline{R_A})(R_{B,t} - \overline{R_B})$$

How to calculate **Sample Covariance** (3 steps):

1. Deviations: Find how far each stock's yearly return was from its own historical average ($R_t - \bar{R}$).
2. The product: Multiply Stock A's deviation by Stock B's deviation for each year, the add them all up.
3. Dividing: Divide the total sum by $\mathbf{n - 1}$ (degree of freedom).

Why?

# Recap: Covariance and correlation

Formula:

$$Corr(R_A, R_B) = \frac{Cov(R_A, R_B)}{SD(R_A)SD(R_B)}$$

- Dividing the co-movement (covariance) by the the individual risk (SD, standard deviations).

Interpreting the number:

- +1 (perfect positive): Move exactly together.
- -1 (perfect negative): Move exactly opposite.
- 0 (uncorrelated): No relationship at all.

# Recap: Covariance and correlation

- SD's formula: Measures of standalone risk of a single asset.

$$SD(R_A) = \sqrt{\frac{\sum(R_{A,t} - \overline{R_A})^2}{n-1}}$$

- SD is simply the square root of Variance.

# 6-3

3. Using the data in the following table, estimate (**a**) the average return and volatility for each stock; (**b**) the covariance between the stocks and (**c**) the correlation between these two stocks.

| Year | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|
| Stock A | − 6% | 7% | 3% | − 8% | 3% | 9% |
| Stock B | 17% | 37% | 12% | − 7% | − 5% | 32% |

**a.** Estimate the average return for each stock.
**b.** Estimate the covariance between the stocks.
**c.** Estimate the correlation between these two stocks.

**a.** Estimate the average return for each stock.

The average return of stock A is _____%. (Round to two decimal places.)

The average return of stock B is _____%. (Round to two decimal places.)

# 6-3 solution

**a.**

Formula:

$$\bar{R}_i = \frac{1}{n}\sum_t R_{i,t}$$

Stock A calculation:

- $\frac{1}{6}\times(-6\% + 7\% + 3\% - 8\% + 3\% + 9\%) = 1.33\%$

Stock B calculation:

- $\frac{1}{6}\times(17\% + 37\% + 12\% - 7\% - 5\% + 32\%) = 14.33\%$

# 6-3 (cont.)

**Match the <u>unit</u> the question asks for!**

**b.** Estimate the covariance between the stocks.

The covariance is _____. (Round to five decimal places.)

**c.** Estimate the correlation between these two stocks.

The correlation is _____. (Round to five decimal places.)

# 6-3 solution

| Year | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|
| Stock A | −6% | 7% | 3% | −8% | 3% | 9% |
| Stock B | 17% | 37% | 12% | −7% | −5% | 32% |

$\overline{R_A} = 1.33\%; \overline{R_B} = 14.33\%$

**b.**

Formula:

$$Cov(R_A, R_B) = \frac{1}{n-1}\sum_t (R_{A,t} - \overline{R_A})(R_{B,t} - \overline{R_B})$$

Calculation:

- $Cov(R_A, R_B) = \frac{1}{6-1} \times [(-0.06 - \mathbf{0.0133}) \times (0.17 - \mathbf{0.1433}) + (0.07 - 0.0133) \times (0.37 - 0.1433) + \cdots + (0.09 - 0.0133) \times (0.32 - 0.1433)] = 0.00815$

# 6-3 solution

**c.**

Formula:

$$Corr(R_A, R_B) = \frac{Cov(R_A, R_B)}{SD(R_A)SD(R_B)}$$

$$SD(R_A) = \sqrt{\frac{\sum(R_A - \overline{R_A})^2}{n-1}}$$

Calculations:

- $SD(R_A) = \sqrt{\frac{(-0.06-0.0133)^2+(0.07-0.0133)^2+\cdots+(0.09-0.0133)^2}{6-1}}$

# 6-3 solution

- $SD(R_B) = \sqrt{\frac{(0.17-0.1433)^2+(0.37-0.1433)^2+\cdots+(0.32-0.1433)^2}{6-1}}$
- $Corr(R_A, R_B) = 0.64753$

Alternative approach: the Excel functions

- COVARIANCE.S (array1, array2)
- STDEV.S (number1, number2…)
- CORREL (array1, array2)

# Recap: portfolio weight

What is a portfolio weight? ($w_i$)?

- It is simply the fraction of your total investment held in one specific asset.

The golden rule:

- No matter how many assets you hold, your portfolio weights **MUST** always add up to **1**.

Formula:

$$w_i = \frac{Value\ of\ Stock\ i}{Total\ Value\ of\ Portfolio}$$

# Recap: portfolio weight

The goal: to find weights, you must first find the total value.

- Scenario A: You are buying stocks (*How much is your position worth?)*

$$Value\ of\ stock\ i = Number\ of\ shares \times Current\ price$$

$$Number\ of\ shares = \frac{Value\ of\ stock\ i}{Current\ price} = \frac{Total\ Value\ of\ Portfolio \times w_i}{Current\ price}$$

- Scenario B: You are looking at a whole company (*How much is the entire company worth?*)

$$Market\ Cap = Total\ shares\ outstanding \times Current\ price$$

# Recap: portfolio weight

Warning: Weight are **not static**!

- If you buy and hold, your weights will drift over time as stock prices change.
- For example, if Stock A goes up and Stock B goes down, then Stock A naturally becomes a larger % of your portfolio.

How to calculate "New Weights"? (the 3-step recipe)

1. Calculate the NEW value of each individual stock.
2. Add them up to find the NEW total portfolio value.
3. Divide Step 1 by Step 2.

# 6-1

1. You are considering how to invest part of your retirement savings.You have decided to put $200,000 into three stocks: 60% of the money in Goldfinger (currently $27/share), 20% of the money in Moosehead (currently $70/share), and the remainder in Venture Associates (currently $8/share). If Goldfinger stock goes up to $38/share, Moosehead stock drops to $66/share, and Venture Associates stock rises to $10 per share.
   **a.** What is the new value of the portfolio?
   **b.** What return did the portfolio earn?
   **c.** If you don't buy or sell any shares after the price change, what are your new portfolio weights?

*Review Only*

**Click the icon to see the Worked Solution (Formula Solution).**

**a.** What is the new value of the portfolio?

The number of shares of Goldfinger is _____ shares. (Round to the nearest integer.)

The number of shares of Moosehead is _____ shares. (Round to the nearest integer.)

The number of shares of Venture is _____ shares. (Round to the nearest integer.)

The new value of the portfolio is $_____. (Round to the nearest dollar.)

# 6-1 solution

Set-ups:

- Initial portfolio value: $200,000
- Asset weights
  - Goldfinger: 60%; Moosehead: 20%; Venture: Remaining 20%

Thus, Initial dollars invested:

- Goldfinger: $\$200{,}000 \times 0.60 = \$120{,}000$
- Moosehead: $\$200{,}000 \times 0.20 = \$40{,}000$
- Venture: $\$200{,}000 \times 0.20 = \$40{,}000$

# 6-1 solution

**a.**

Formula:

$$Number\ of\ shares = \frac{Total\ Value\ of\ Portfolio \times w_i}{Current\ price}$$

Let $\mathrm{N}(i)$ be the number of share in stock *i*, then

- $N_{Goldfinger} = \frac{120{,}000}{27} = 4{,}444$
- $N_{Moosehead} = \frac{40{,}000}{70} = 671$
- $N_{Venture} = \frac{40{,}000}{8} = 5{,}000$

# 6-1 solution

Now, we look at the value after the stock prices change.

Step 1:

- Goldfinger (new price \$38): $4{,}444 \times \$38 = \$168{,}872$
- Moosehead (new price \$66): $571 \times \$66 = \$37{,}686$
- Venture (new price \$10): $5{,}000 \times \$10 = \$50{,}000$

Step 2: The total new portfolio value:

- $\$168{,}872 + \$37{,}686 + \$50{,}000 = \$256{,}558$

Step 3 → “New Weights” (question c)

# 6-1 (cont.)

**b.** What return did the portfolio earn?

The portfolio earned a return of _____%. (Round to two decimal places.)

**c.** If you don't buy or sell any shares after the price change, what are your new portfolio weights?

Weight of Goldfinger is now _____%. (Round to two decimal places.)

Weight of Moosehead is now _____%. (Round to two decimal places.)

Weight of Venture is now _____%. (Round to two decimal places.)

# 6-1 solution (cont.)

**b.**

Formula:

$$R_p = \frac{New\ value - Initial\ value}{Initial\ value}$$

Calculation:

- $R_p = \frac{\$256{,}558 - \$200{,}000}{\$200{,}000} = 0.28279$
- Answer: The portfolio earned a return of **28.28%**

# 6-1 solution (cont.)

**c.**

Formula:

$$w_i = \frac{Value\ of\ Stock\ i}{Total\ Value\ of\ Portfolio}$$

Step 3: Calculations ("New Weights"):

- $w_{Goldfinger} = \frac{\$168,872}{\$256,558} = 65.82\%$
- $w_{Moosehead} = \frac{\$37,686}{\$256,558} = 14.69\%$
- $w_{Venture} = \frac{\%50,000}{\$256,558} = 19.49\%$

# Recap: portfolio return & risk

**Method 1: Ex-post approach**

- Treat the entire portfolio as a <u>single, standalone</u> asset.
- Use this when you have a table of either the portfolio's total value **history** OR the individual history of each asset plus their weights.

Calculating return:

$$R_p = \frac{New\ value - Initial\ value}{Initial\ value}, \qquad \overline{R_P} = \frac{1}{n}\sum_t R_{p,t}$$

Calculating risk:

$$SD_p = \sigma_p = \sqrt{\frac{\sum(R_{p,t} - \overline{R_p})^2}{n-1}}$$

# Recap: portfolio return & risk

**Method 2: Ex-ante approach**

- Build the portfolio from its <u>individual</u> assets and weights.
- Used when you are given a "**forecast**" that provides the final parameters (e.g., expected return, $E(R_i)$) but **NOT** the raw history.

Calculating return:

$$E(R_p) = \sum w_i E(R_i)$$

$$\rightarrow E(R_p) = w \times E(R_A) + w \times E(R_B) + w \times E(R_C), \qquad if\ N = 3$$

# Recap: portfolio return & risk

$$\rightarrow E\left(R_p\right) = w \times E(R_A) + (1 - w) \times E(R_B), \qquad if\ N = 2$$

Calculating risk (two-asset portfolio):

$$\sigma_P = \sqrt{w^2\sigma_A^2 + 2w(1 - w)\sigma_A\sigma_B\rho_{AB} + (1 - w)^2\sigma_B^2}$$

# 6-4

**Ex-post approach**

4. Using the data in the following table consider a portfolio that maintains a 55% weight in stock A and a 45% weight in stock B.

| Year | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|
| Stock A | − 15% | 5% | 5% | − 3% | 4% | 14% |
| Stock B | 15% | 10% | 19% | − 6% | − 12% | 30% |

**a.** What is the return each year of this portfolio?
**b.** Based on your results from part (**a**), compute the average return and volatility of the portfolio.

**a.** What is the return each year of this portfolio?

Enter the return of this portfolio for each year in the table below: (Round to two decimal places.)

| Year | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|
| Portfolio | _____ % | _____ % | _____ % | _____ % | _____ % | _____ % |

**b.** Based on your results from part (**a**), compute the average return and volatility of the portfolio.

The average return of the portfolio is _____ %. (Round to two decimal places.)

The volatility of the portfolio is _____ %. (Round to two decimal places.)

# 6-4 solution

| Year | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|---|---|
| Stock A | −15% | 5% | 5% | −3% | 4% | 14% |
| Stock B | 15% | 10% | 19% | −6% | −12% | 30% |

**a.**

Formula:

$$R_p = w \times R_A + (1 - w) \times R_B$$

2004: $0.55 \times (-15\%) + 0.45 \times 15\% = -1.50\%$

2005: $0.55 \times 5\% + 0.45 \times 10\% = 7.25\%$

2006: $0.55 \times 5\% + 0.45 \times 19\% = 11.30\%$

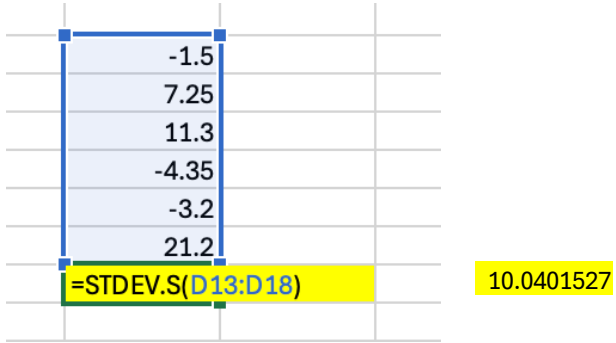
2007: $0.55 \times (-3\%) + 0.45 \times (-6\%) = -4.35\%$

2008: $0.55 \times 4\% + 0.45 \times (-12\%) = -3.20\%$

2009: $0.55 \times 14\% + 0.45 \times 30\% = 21.20\%$

# 6-4 solution

Alternative approach:
Excel function




**b.**

$$\overline{R_P} = \frac{1}{n}\sum_t R_{p,t}$$

- $\overline{R_p} = \frac{1}{6} \times (-1.50\% + 7.25\% + 11.30\% - 4.35\% - 3.20\% + 21.20\%) = 5.12\%$

Formula:

$$SD_p = \sigma_p = \sqrt{\frac{\sum(R_{p,t} - \bar{R}_p)^2}{n-1}}$$

- $\sigma_p = \sqrt{\frac{(-1.50\% - 5.12\%)^2 + \cdots + (21.20\% - 5.12\%)^2}{6-1}} = 10.04\%$

# 6-2

**Ex-ante approach**

2. Consider a world that only consists of the three stocks shown in the following table:

| Stock | Total Number of Shares Outstanding | Current Price per Share | Expected Return |
|---|---|---|---|
| First Bank | 114 million | $107 | 16% |
| Fast Mover | 55 million | $120 | 10% |
| Funny Bone | 198 million | $26 | 14% |

**a.** Calculate the total value of all shares outstanding currently.
**b.** What fraction of the total value outstanding does each stock make up?
**c.** You hold the market portfolio, that is, you have picked portfolio weights equal to the answer to part b with each stock's weight is equal to its contribution to the fraction of the total value of all stocks. What is the expected return of your portfolio?

**a.** Calculate the total value of all shares outstanding currently.

The total value of all shares outstanding currently is $_____ million. (Round to the nearest integer.)

# 6-2 solution

| Stock | Total Number of Shares Outstanding | Current Price per Share |
|---|---|---|
| First Bank | 114 million | $107 |
| Fast Mover | 55 million | $120 |
| Funny Bone | 198 million | $26 |

**a.**

Formula:

$$Market\ Cap = Total\ Share\ Outstanding \times Price\ per\ Share$$

Individual Calculations:

- First Bank: $114\ million \times \$107 = \$12{,}198\ million$
- Fast Mover: $55\ million \times \$120 = \$6{,}600\ million$
- Funny Bone: $198\ million \times \$26 = \$5{,}148\ million$

Total Market Value: $\$12{,}198 + \$6{,}600 + \$5{,}148 = \$23{,}946\ million$

# 6-2 solution

**b.**

Formula:

$$w_i = \frac{The\ value\ of\ Stock\ i}{The\ Portfolio\ Value}$$

Calculations:

- $w_{FirstBank} = \frac{\$12,198}{\$23,946} = 50.94\%$
- $w_{FastMover} = \frac{\$6,600}{\$23,946} = 27.56\%$
- $w_{FunnyBone} = \frac{\$5,148}{\$23,946} = 21.50\%$

# 6-2 solution

| Stock | Expected Return |
|---|---|
| First Bank | 16% |
| Fast Mover | 10% |
| Funny Bone | 14% |

**c.**

Formula:

$$E(R_p) = \sum w_i E(R_i)$$

$$\rightarrow E(R_p) = (w_1 \times E(R_A)) + (w_2 \times E(R_B)) + (w_3 \times E(R_C))$$

Calculations:

- First Bank Contribution: $50.94\% \times 16\% = 0.081504$
- Fast Mover Contribution: $27.56\% \times 10\% = 0.027560$
- Funny Bone Contribution: $21.50\% \times 14\% = 0.030100$
- Answer: $8.1504\% + 2.656\% + 3.01\% = \mathbf{13.92}\%$

# 6-6

**Ex-ante approach**

6. Suppose Intel's stock has an expected return of 20.0% and a volatility of 3.0%, while Coca-Cola's has an expected return of 7.0% and volatility of 3.0%. If these two stocks were perfectly negatively correlated (i.e., their correlation coefficient is $-1$),
**a.** Calculate the portfolio weights that remove all risk.
**b.** If there are no arbitrage opportunities, what is the risk-free rate of interest in this economy?

$\rightarrow \rho_{AB} = -1$

**a.** Calculate the portfolio weights that remove all risk. $\rightarrow \sigma_p = 0$

The portfolio weight of Intel would be _____%. (Round to two decimal places.)

The portfolio weight of Coca-Cola would be _____%. (Round to two decimal places.)

**b.** If there are no arbitrage opportunities, what is the risk-free rate of interest in this economy?

The risk-free rate of interest in this economy is _____%. (Round to two decimal places.)

ID: P11-22 (similar to)

# 6-6 solution

**a.**

Formula:

$$\sigma_P = \sqrt{w^2\sigma_A^2 + 2w(1-w)\sigma_A\sigma_B\rho_{AB} + (1-w)^2\sigma_B^2}$$

- If we plug $\rho_{xy} = -1$ into the formula, the formula above is a "perfect square trinomial" (like $a^2 - 2ab + b^2 = (a-b)^2$).
- We then can compress the formula significantly:

$$\sigma_P = \sqrt{(w\sigma_A - (1-w)\sigma_B)^2}$$
$$\sigma_P = |w\sigma_A - (1-w)\sigma_B|$$

# 6-6 solution

- Our goal is to "remove all risk", which means we want the portfolio SD to equal exactly zero ($\sigma_P = 0$).

$$w\sigma_A - (1-w)\sigma_B = 0$$
$$w\sigma_A = (1-w)\sigma_B$$

- Now plug in the data of the volatility ($\sigma_A = 3\%, \sigma_B = 3\%$)

$$w \times 3\% = (1-w) \times 3\%$$

Thus

$$w = \frac{100\%}{2} = 50\%$$

# 6-6 solution

**b.**

- **"no arbitrage" principle:** If a portfolio has absolutely zero risk, its return must <u>exactly equal</u> the market's "risk-free rate" ($r_f$).

Formula:

$$E(R_p) = \sum_i w_i E(P_i)$$

- $E(R_p) = 50\% \times 20\% + 50\% \times 7\% = 13.50\%$
- Since the portfolio has no risk, the risk free interest is also 13.50%.

# 6-7

$\sigma_f = 0$ **(risk-free rate)!**

7. Suppose you have $325,000 in cash, and you decide to borrow another $58,500 at a 7% interest rate to invest in the stock market. You invest the entire $383,500 in a portfolio J with a 17% expected return and a 24% volatility.
**a.** What is the expected return and volatility (standard deviation) of your investment?
**b.** What is your realized return if J goes up 25% over the year?
**c.** What return do you realize if J falls by 32% over the year?

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**a.** What is the expected return and volatility (standard deviation) of your investment?

The expected return of your investment is _____%. (Round to two decimal places.)

The volatility (standard deviation) of your investment is _____%. (Round to two decimal places.)

**b.** What is your realized return if J goes up 25% over the year?

Your realized return if J goes up 25% over the year is _____%. (Round to two decimal places.)

**c.** What return do you realize if J falls by 32% over the year?

The return you realize if J falls by 32% over the year is _____%. (Round to two decimal places.)

# 6-7 solution

**a.**

Formula:

$$E(R_p) = (w_J \times E(R_J)) + ((1 - w_J) \times r_f)$$

- $E(R_p) = 1.18 \times 17\% + (1 - 1.18) \times 7\% = 18.80\%$

# 6-7 solution

Formula:

$$\sigma_P = \sqrt{w_J^2\sigma_J^2 + 2w_J(1-w_J)\sigma_J\sigma_f\rho_{Jf} + (1-w_J)^2\sigma_f^2}$$

Since

$$\sigma_f = 0$$

**Zero times anything is zero!**

$$\sigma_p = |w_J \times \sigma_J|$$

- $\sigma_p = 1.18 \times 24\% = 28.32\%$

# 6-7 solution

**b.**

Formula:

$$R_p = (w_J \times \boldsymbol{R_J}) + ((1 - w_J) \times r_f)$$

- $E(R_p) = 1.18 \times 25\% + (1 - 1.18) \times 7\% = 28.24\%$

**c.**

- $E(R_p) = 1.18 \times (-32\%) + (1 - 1.18) \times 7\% = -39.02\%$